

Challenge Problems Week 5

Two Unrelated Problems

UCQ Education Team

Problem 1 Take the four-qubit state

$$\left(\frac{1}{\sqrt{2}} |00\rangle_{12} + \frac{1}{\sqrt{2}} |11\rangle_{12} \right) \otimes \left(\frac{1}{\sqrt{2}} |00\rangle_{34} + \frac{1}{\sqrt{2}} |11\rangle_{34} \right)$$

Suppose one chose to entangle qubits 2 and 3 together by performing a two-qubit measurement in the Bell basis, and they ended up measuring

$$|\mathbb{1}\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23}),$$

Show that this necessarily leads to an entangled Bell pair state of qubits 1 and 4. This type of phenomenon is referred to as the *monogamy of entanglement*.

Problem 2. Since the Grover iterations never land exactly at an angle of $\theta = \frac{\pi}{2}$, Grover's search is inherently *probabilistic*. As $N \rightarrow \infty$, the probability of measuring the correct answer approaches 1. What is the leading-order correction to this for large N ?